The Derivative as a Limit and an Instantaneous Rate Answer Key

AP Calculus AB / BC

Quick Answers

1. 8
2. 6
3. $3t^2 - 12t + 9$
4. 24
5. 75
6. -0.25
7. 6
8. —

Common wrong answers

2. *If they got 7: reported the average rate over the whole interval from 2 to 3 instead of the limiting value*
4. *If they got 20: evaluated the original volume function instead of its derivative*
7. *If they got 12: subtracted the two distances but never divided by the change in time*

How to use this key. On this sheet the number is rarely the point. Credit the two moves that are: simplifying the difference quotient *before* taking the limit (so the 0/0 form is resolved rather than evaluated), and stating what the resulting number measures, with its units. Problem 8 is open reasoning and is flagged for manual review — a model response is given below to compare against, not a key to match word for word.

1. $f(x) = x^2$; complete the table and find $f'(4)$.
Solution: The difference quotient simplifies once and for all:

$$\frac{(4+h)^2 - 4^2}{h} = \frac{16 + 8h + h^2 - 16}{h} = \frac{8h + h^2}{h} = 8 + h \quad (h \neq 0).$$

So the table entries are 9, 8.5, 8.1, 8.01 — each is $8+h$, and the column is visibly closing on 8 from above. Because the simplified form is $8 + h$, the limit is immediate: $\boxed{8}$. Note that cancelling the h is legitimate only because the limit never uses $h = 0$ itself, only values near it.

2. $s(t) = t^2 + 2t$; average velocities from $t = 2$, then the instantaneous velocity there.
Solution: $s(2) = 8$, and

$$\frac{s(2+h) - s(2)}{h} = \frac{(2+h)^2 + 2(2+h) - 8}{h} = \frac{6h + h^2}{h} = 6 + h.$$

The table reads 7, 6.5, 6.1, 6.01. Each of those is a genuine average velocity over a real window, and none of them is the instantaneous velocity; the instantaneous velocity is what they approach: $\boxed{\text{6 meters per second}}$. A student who stops at the first column and reports 7 has reported the average velocity from $t = 2$ to $t = 3$, which is the classic confusion this problem is built to expose.

3. $V(t) = t^3 - 6t^2 + 9t$; find $V'(t)$ and say what it measures.

Solution: Differentiating term by term, $\boxed{3t^2 - 12t + 9}$.
Meaning: $V'(t)$ is the instantaneous rate at which the volume in the tank is changing at time t — how fast gallons are being gained (positive) or lost (negative) at that instant, not over any interval. It is a function of t, so the tank's rate is different at different moments.

4. Find the instantaneous rate at $t = 5$.

Solution: $V'(5) = 3(25) - 12(5) + 9 = 75 - 60 + 9 = 24$, so the rate is $\boxed{\text{24 gallons per minute}}$. It is positive, so at that instant the volume is increasing — the tank is filling. Substituting into V instead of V' gives $125 - 150 + 45 = 20$ gallons, which is the amount of water present, not a rate; the units are the tell.

5. Recognise $\lim_{h \to 0} \dfrac{(5+h)^3 - 125}{h}$.

Solution: The quotient has the shape $\frac{f(a+h)-f(a)}{h}$. The $(5 + h)$ shows $a = 5$, and cubing it shows $f(x) = x^3$; the subtracted constant confirms it, since $f(5) = 125$. So the limit is $f'(5)$ for $f(x) = x^3$. Since $f'(x) = 3x^2$, the value is $3(25) = \boxed{75}$. Reading a limit this way turns a hard algebraic simplification into a one-line derivative evaluation, which is exactly why the pattern is worth recognising.

6. $f(x) = 1/x$; find $f'(2)$ from the definition.

Solution:

$$\frac{\frac{1}{2+h} - \frac{1}{2}}{h} = \frac{\frac{2-(2+h)}{2(2+h)}}{h} = \frac{-h}{2h(2+h)} = \frac{-1}{2(2+h)}.$$

Now the $h \to 0$ substitution is safe: the expression becomes $\frac{-1}{4}$, that is $\boxed{-0.25}$. The sign is meaningful — $1/x$ is decreasing at $x = 2$, so its instantaneous rate there must be negative.

7. Table of distances; estimate the velocity at $t = 3$.

Solution: With no formula there is no limit to take, so the best available estimate is the average rate over the smallest window we have that is centred on $t = 3$: from $t = 2$ to $t = 4$,

$$\frac{41 - 29}{4 - 2} = \frac{12}{2} = 6,$$

so about $\boxed{\text{6 meters per second}}$. The centred window is better because the one-sided estimates ($35 - 29 = 6$ over one second forward from 2, and $41 - 35 = 6$ over one second forward from 3) each lean to one side of $t = 3$, while the centred window averages the behaviour just before and just after, cancelling most of the first-order error. Forgetting to divide by the 2-second window gives 12, a distance rather than a rate.

8. Open response — flagged for manual review.
Model response: For $f(x) = mx + k$ the difference quotient is

$$\frac{m(a+h) + k - (ma+k)}{h} = \frac{mh}{h} = m,$$

which contains no h at all. It is already equal to m for every nonzero h and every a, so the limit is m regardless of where you stand: a line has one slope. That is also why a table of average rates for a linear function needs no limit — every entry in the table is already the answer, so nothing is being approached. For a curve the quotient still depends on h, and the limit is what removes that dependence.
Units: in $\dfrac{V(t+h) - V(t)}{h}$ the numerator is a difference of volumes, so it carries gallons, and the denominator is a difference of times, so it carries minutes. The quotient therefore has units of gallons per minute, and taking a limit does not change units — every term in the sequence already had them. This is the general rule: a derivative's units are always the output units divided by the input units.
What to look for when grading: the algebra showing the h cancels and nothing survives; the explicit statement that the result is independent of a; and units derived from the quotient rather than asserted.